

The purpose of this assignment is to provide students with hands-on experience in applying Python programming and data analysis skills to a dataset. By working with a given dataset, students will practice fundamental tasks such as data preprocessing using Python libraries, such as NumPy and pandas. They will also explore methods for analyzing and visualizing data patterns through Matplotlib.

1 Background

The dataset used in this assignment comes from climate downscaling experiments conducted with a deep learning model. Climate downscaling is similar to image super-resolution. In general, the purpose is to enhance the data resolution (or data quality) by a machine learning model. However, the climate downscaling application could derive new output data variables. Domain experts prepare a **test dataset** to evaluate the model prediction performance. The dataset spans **365 days**, with one sample per day, providing a year-long record for analysis.

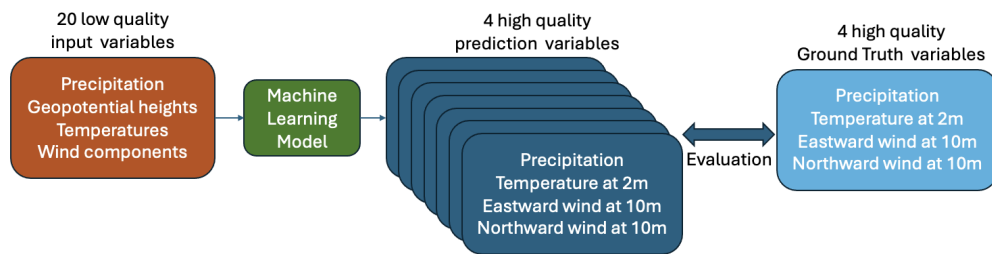


Figure 1: Illustration of one input instance (day) prediction and evaluation.

Each sample contains **20 model input variables** representing atmospheric and surface conditions, including:

- Precipitation
- Geopotential height (500.0, 700.0, 850.0, 925.0 hPa)
- Temperature (500.0, 700.0, 850.0, 925.0 hPa, 2m)
- Eastward and northward wind components (500.0, 700.0, 850.0, 925.0 hPa, 10m)

All input fields are provided at a spatial resolution of **208 × 208**, obtained by interpolating low-resolution data using a naive interpolation method. Therefore, although the model input data resolution is the same as that of the model output, the input data quality is lower.

The **model output variables** consist of four climate variables:

- Precipitation

- Temperature at 2m
- Eastward wind at 10m
- Northward wind at 10m

These outputs are also given at the same **208 × 208 resolution**.

Additionally, the dataset includes the **ground truth output fields**, which serve as reference data for evaluating prediction accuracy. Therefore, the ground truth output fields are the same as those of the model output variables. In addition, the machine learning model is designed to capture prediction uncertainty by generating **64 ensemble members** for each input, meaning that for the same input fields, the model can produce 64 slightly varied predictions. This ensemble approach allows for estimation of prediction error and uncertainty, offering valuable opportunities for statistical analysis and visualization.

If you are interested in more downscaling related background information, you can read this paper <https://arxiv.org/html/2309.15214v3>.

2 Dataset Download Instruction and Dataset Loading Example

The dataset for this assignment is provided as a large NetCDF file (approximately **15 GB**). Students can download the file from the following URL:

https://drive.google.com/file/d/1syw4OExircETM6P7_o_Jk4Suhst2jisg/view?usp=sharing

The dataset is put in a Google Drive folder. Since file size may cause download limitations, please download it as early as possible. If you encounter any issues during download, contact the TA as early as possible.

About NetCDF Files

The dataset is stored in the **NetCDF** (Network Common Data Form, file extension `.nc`) format. NetCDF is a standard format commonly used in the atmospheric and climate sciences for storing multidimensional scientific data, such as time, latitude, longitude, and vertical levels. It is well-suited for large datasets and can be efficiently accessed using Python libraries such as `netCDF4`.

Example Notebook

An example Jupyter Notebook (`.ipynb`) file has been prepared and made available on Moodle. This notebook demonstrates how to load the dataset using the `netCDF4` Python library and visu-

alize basic variables with `Matplotlib`. Students are encouraged to begin with this notebook to familiarize themselves with the data structure and plotting procedures.

Additional File: Land Mask

Alongside the main dataset, students should also download another NetCDF file:

<https://drive.google.com/file/d/1yjmTtzdhjjdNekxvrCHhDeVZnppFCdor/view?usp=sharing>

This file contains a **208 × 208 land mask**, where land grid cells are marked as 1 and ocean grid cells as 0. Since model performance is often evaluated only over land areas, this file can be used to filter and restrict analysis accordingly.

Dataset Structure

The main dataset file contains three groups of variables:

- **truth**: The ground-truth reference data of high-resolution (quality) ground truth data.
- **prediction**: The outputs generated by the deep learning downscaling model. Each sample contains **64 ensemble members**, providing slightly varied predictions to capture uncertainty.
- **input**: The original input variables provided to the model.

Research Questions for Visualization Analysis

In scientific studies and machine learning evaluations, data visualization plays a critical role in interpreting model performance and identifying meaningful physical patterns. For this assignment, you are expected to act as an atmospheric data analyst, and address the following questions using effective and well-designed Python visualizations (You are supposed to use `Matplotlib` library). Your visualizations should provide clear evidence to help a climate expert understand the answers derived from the dataset.

1. Is the value distribution of the **output precipitation variable** similar to that of the **ground-truth precipitation variable** over the whole year?
2. Is the value distribution of the **input precipitation variable** similar to that of the **ground-truth precipitation variable** over the whole year?

3. Is the value distribution of the **output temperature variable** similar to that of the **ground-truth temperature variable** over the whole year?
4. Is the value distribution of the **input temperature variable** similar to that of the **ground-truth temperature variable** over the whole year?
5. What are the distributions of **Mean Absolute Error (MAE)** for temperature variables and for precipitation variables?
6. Does increasing the number of ensemble members improve the average prediction accuracy (measured by MAE)? Do all four output variables exhibit the same trend, or are there differences among them?
7. Are there any notable patterns in prediction error across different **months or seasons**? If evaluated by MAE and CRPS (Continuous Ranked Probability Score), do the patterns differ?
8. Does the **3-sigma uncertainty range** adequately cover the ground truth? Are the conclusions consistent across all variables?
9. In different months, do the spatial patterns of **high precipitation error areas** differ?
10. If we only consider the top 10% of the highest precipitation events, do the spatial patterns of high precipitation error areas differ across months?

Note that you should answer the above questions by the data points only on the land area.

Assignment Instructions

You are required to submit your answers in the form of a Jupyter Notebook (.ipynb). Each question should be addressed in a separate code cell (notebook block). In each block, you are expected to:

- At the beginning of each notebook block, please include the corresponding question again as a Python comment.
- At the beginning of each notebook block, add short explanatory comments to answer your conclusion and describe your data analysis and visualization approach/strategy and highlight the evidence shown in the visualization.
- Provide Python code that performs the necessary data processing and visualization.
- Create clear, effective, and properly labeled visualizations that can address the question.
- You should place both the main NetCDF file and the land mask file in the same folder as your .ipynb file. When accessing them in your program, use only relative file paths, not absolute paths.

Submission Guidelines

- **Deadline:** October 4, 2026 at 11:59 PM.
- This is an **individual assignment**. Every student must complete and submit their own work.
- Submit your Jupyter Notebook file (.ipynb) to Moodle. No other file formats are required.
- Please name your notebook file as your **Student ID** (e.g., 410123456.ipynb). Failure to follow the naming convention will result in a deduction of points.

Contact Information

- **TA Email:** 61447071S@gapps.ntnu.edu.tw 61347081s@gapps.ntnu.edu.tw
- **Instructor Email:** kcwang@ntnu.edu.tw

For any questions related to this homework, please send your inquiry to **both** the TA and the course instructor by email.